Unified Modeling Language (UML)

What is UML?

- The Unified Modeling Language (UML) is a graphical language for visualizing, specifying, constructing, and documenting the artifacts of a software-intensive system.

Goals of UML?

- Provide users with a ready-to-use, expressive visual modeling language so they can develop and exchange meaningful models.

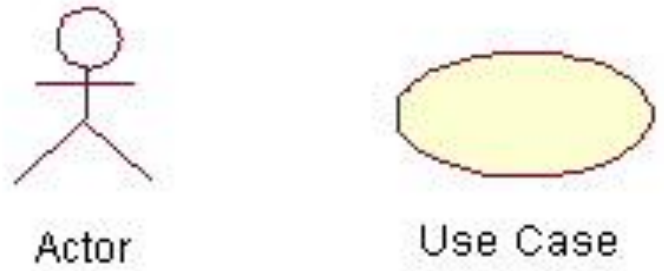
UML Diagrams

1) **Use Case Diagram .**
2) **Activity Diagram .**
3) **Class Diagram .**
4) **Communication Diagram .**
5) **Component Diagram .**
6) **Deployment Diagram .**
7) **Object Diagram .**
8) **Package Diagram .**
9) **Sequence Diagram .**
10) **State Diagram .**

Use Case Diagram

- A use case is a set of scenarios that describing an interaction between a **user** and a **system**. It displays the relationship among actors and use cases.

- The two main components of a use case diagram are **use cases** and **actors**



Use Case Diagram

- **Use Case**

  A use case describes a sequence of actions that provide something of measurable value to an actor .



- **Actor**

  An actor is a person, organization, or external system that plays a role in one or more interactions with your system



Identifying Use Cases

- Which functions does the actor require from system?
- Does the actor need to read, create, destroy, modify, or store some kind of information in the system?
- Does the actor have to be notified about events in the system

Identifying actors

- Useful questions
  - **Who will use the main functionality of the system(primary actors)?**
  - **Who will need to maintain, administrate, and keep the system working (secondary actors)?**
  - **Which hardware devices does the system need to handle?**
  - **With which other systems does the system need to interact?**
  - **Who or what has an interest in the results (the value) that the system produces?**

Use case Examples

Bank System

- A bank has several automated teller machines (ATMs), which are geographically distributed and connected via a wide area network to a central server.

- Each ATM machine has a
  - **Card reader**
  - **Cash dispenser**
  - **keyboard/display**
  - **Receipt printer.**

- By using the ATM machine, a customer can
  - **Withdraw cash from either a checking or savings account**
  - **Query the balance of an account**
  - **Transfer funds from one account to another.**

Bank System

- A transaction is initiated when a customer inserts an ATM card into the card reader.
- Encoded on the magnetic strip on the back of the ATM card are the **card number**, the **start date**, and the **expiration date.**
- Assuming the card is recognized, the system validates the ATM card to determine that
  - **The expiration date has not passed,**
  - **The user-entered PIN (personal identification number) matches the PIN maintained by the system**
  - **The card is not lost or stolen.**
- The customer is allowed three attempts to enter the correct PIN; the card is confiscated if the third attempt fails.
- Cards that have been reported lost or stolen are also confiscated.

Bank System

- If the PIN is validated satisfactorily, the customer is prompted for a withdrawal, query, or transfer transaction.

- Before a withdrawal transaction can be approved, the system, determines that
  - **Sufficient funds exist in the requested account,**
  - **The maximum daily limit will not be exceeded**
  - **There are sufficient funds at the local cash dispenser.**

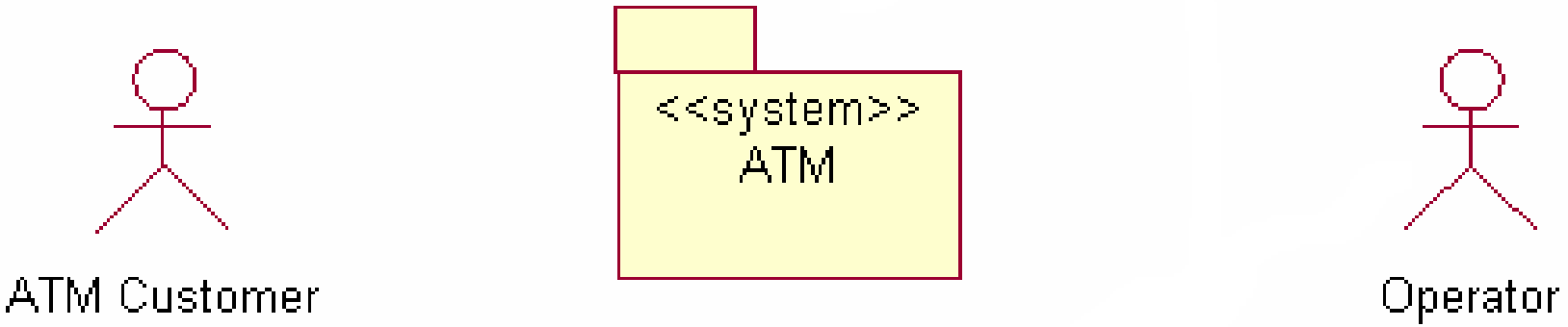
- If the transaction is approved, the requested amount of cash is dispensed, a receipt is printed containing information about the transaction, and the card is ejected.

- Before a transfer transaction can be approved, the system determines that the customer has at least two accounts and that there are sufficient funds in the account to be debited

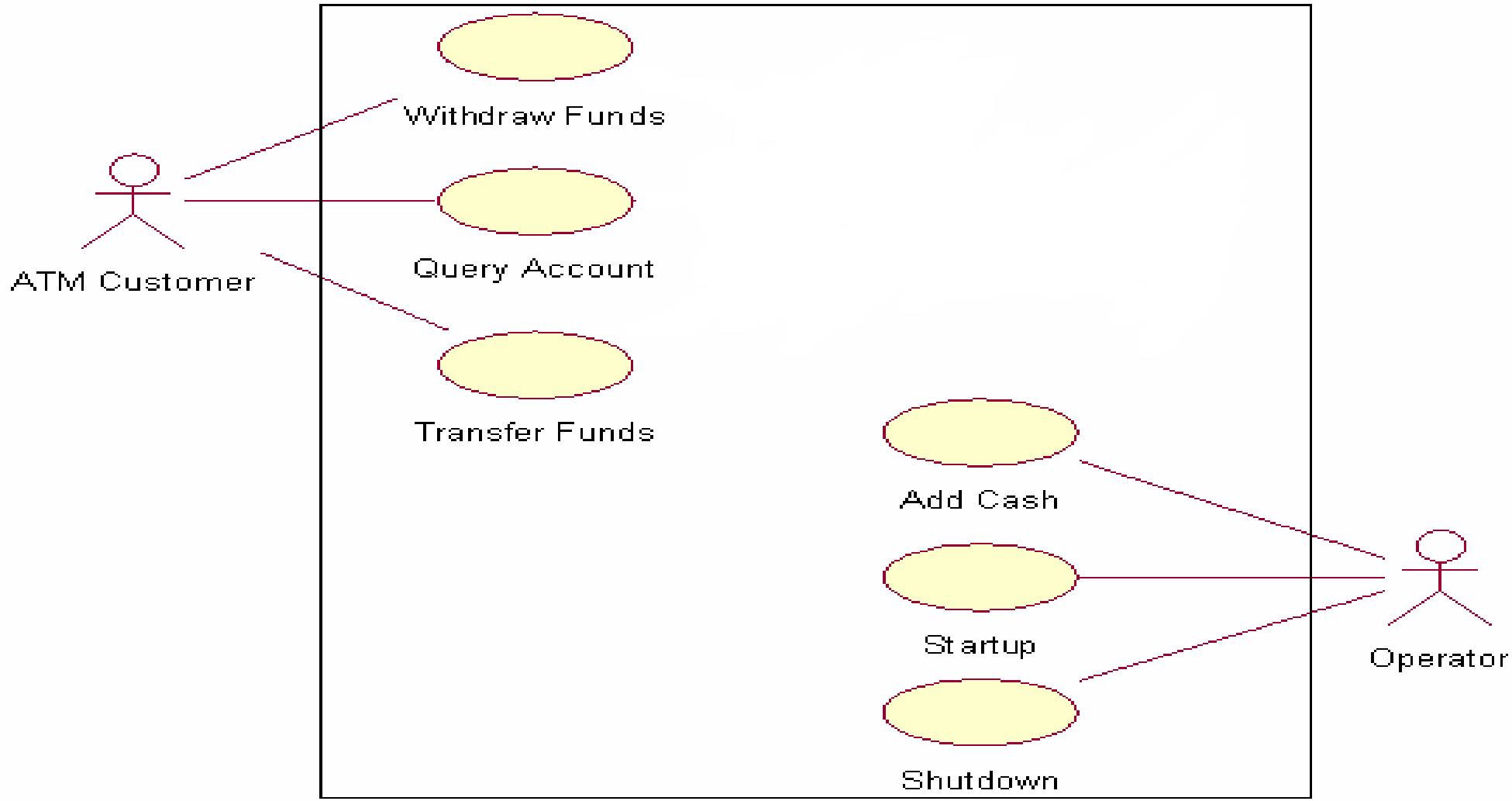
Bank System

- **For approved query and transfer requests, a receipt is printed and the card is ejected.**
- **A customer many cancel a transaction at any time; the transaction is terminated and the card is ejected.**
- **Customer records, account records, and debit card records are all maintained at the server.**
- **An ATM operator may start up and close down the ATM to replenish the ATM cash dispenser and for routine maintenance.**

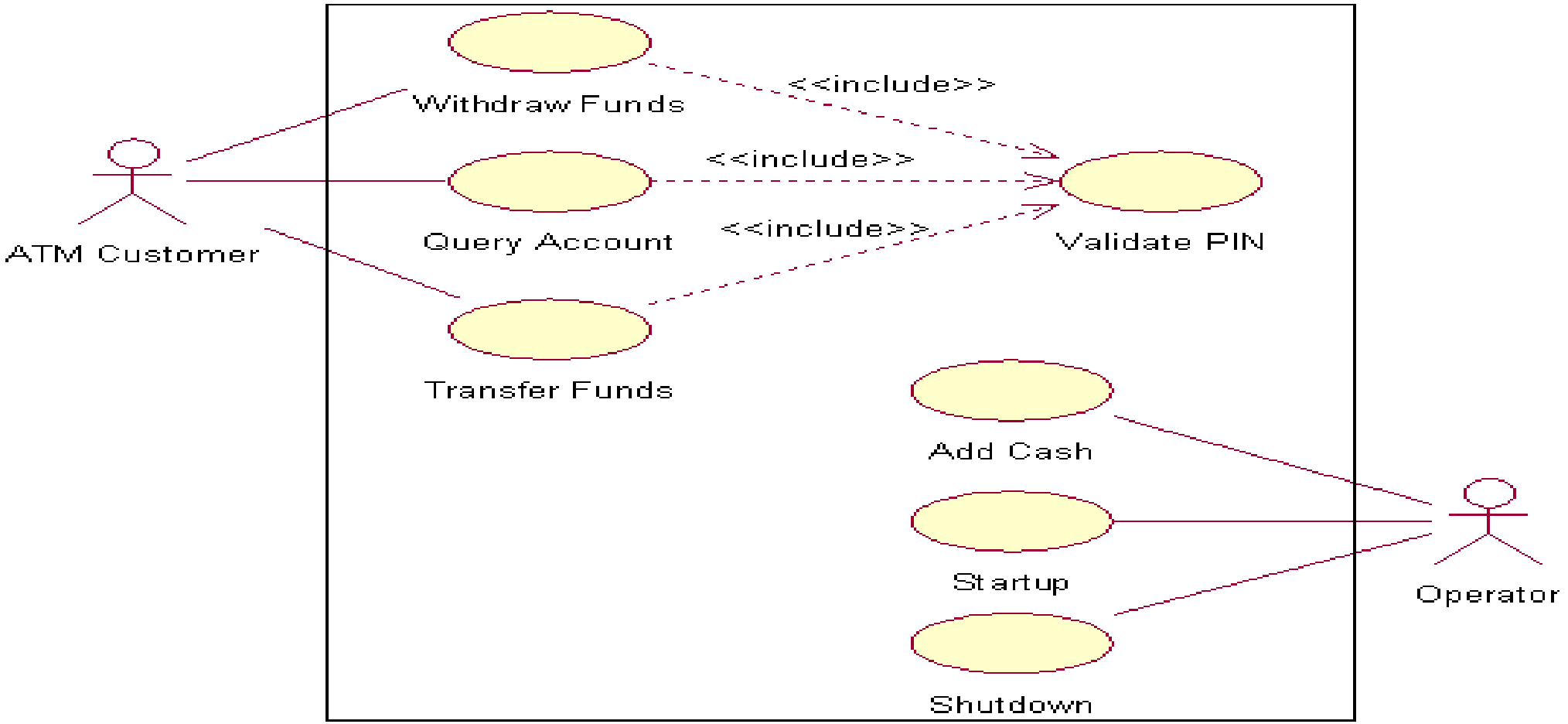
ATM System



ATM: Use Case Model

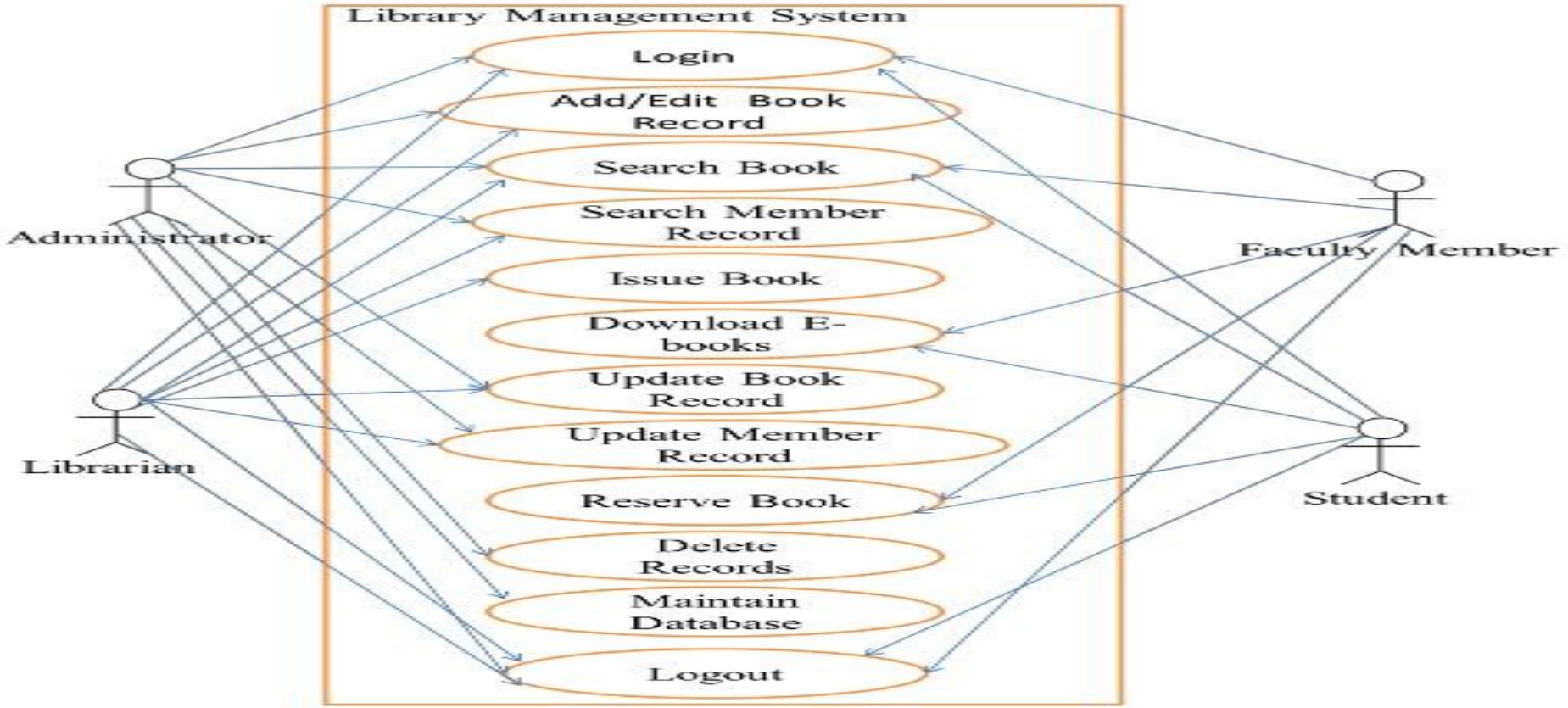


ATM: Use Case Model



Library Management System use case

3.2 Use case diagram



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Use cases (with descriptions and details)

- **Library Management System Description :**

The purpose of the system is to allow for storing details of a large number of books and allow for add, search, borrow, return facilities separately to administrator , staff and students.

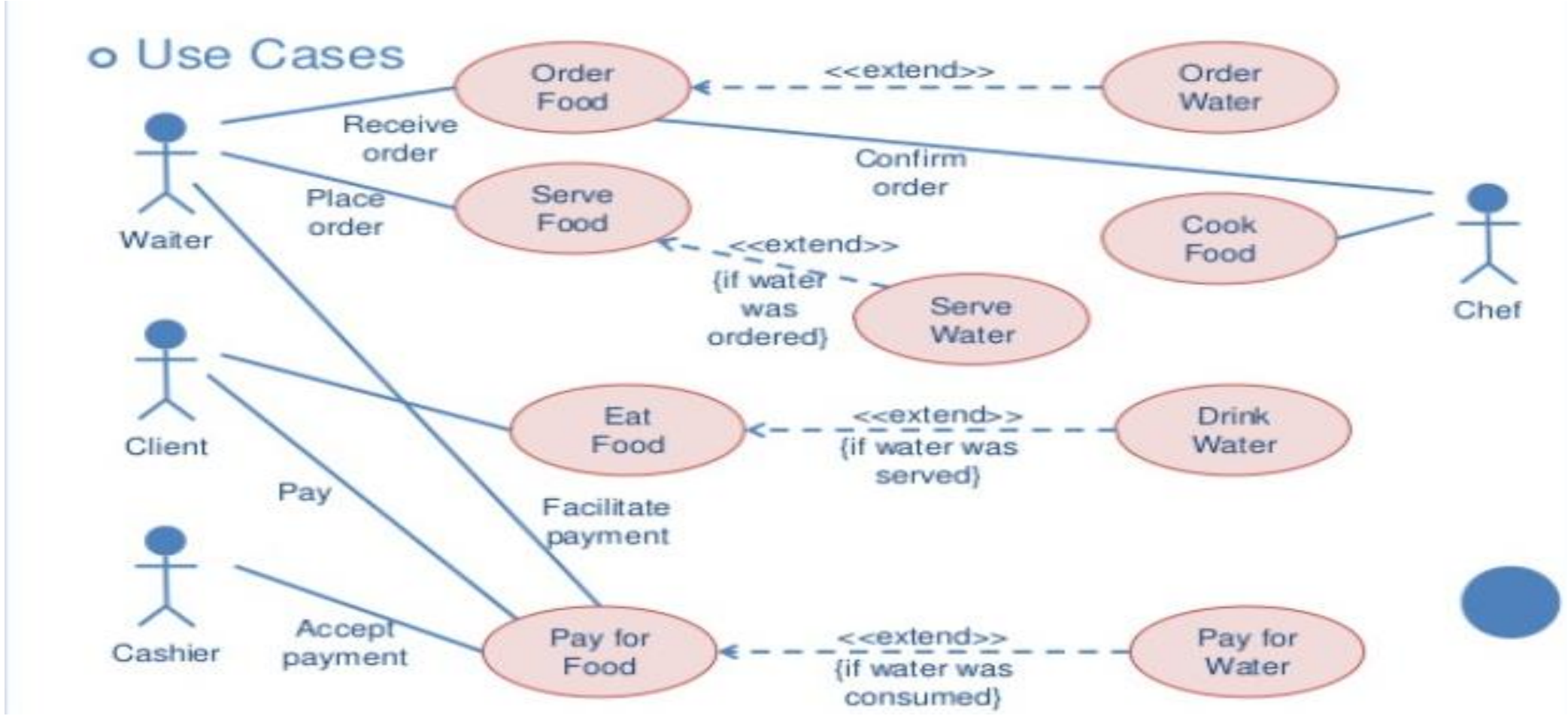
For example :

explain (Search) use cases .

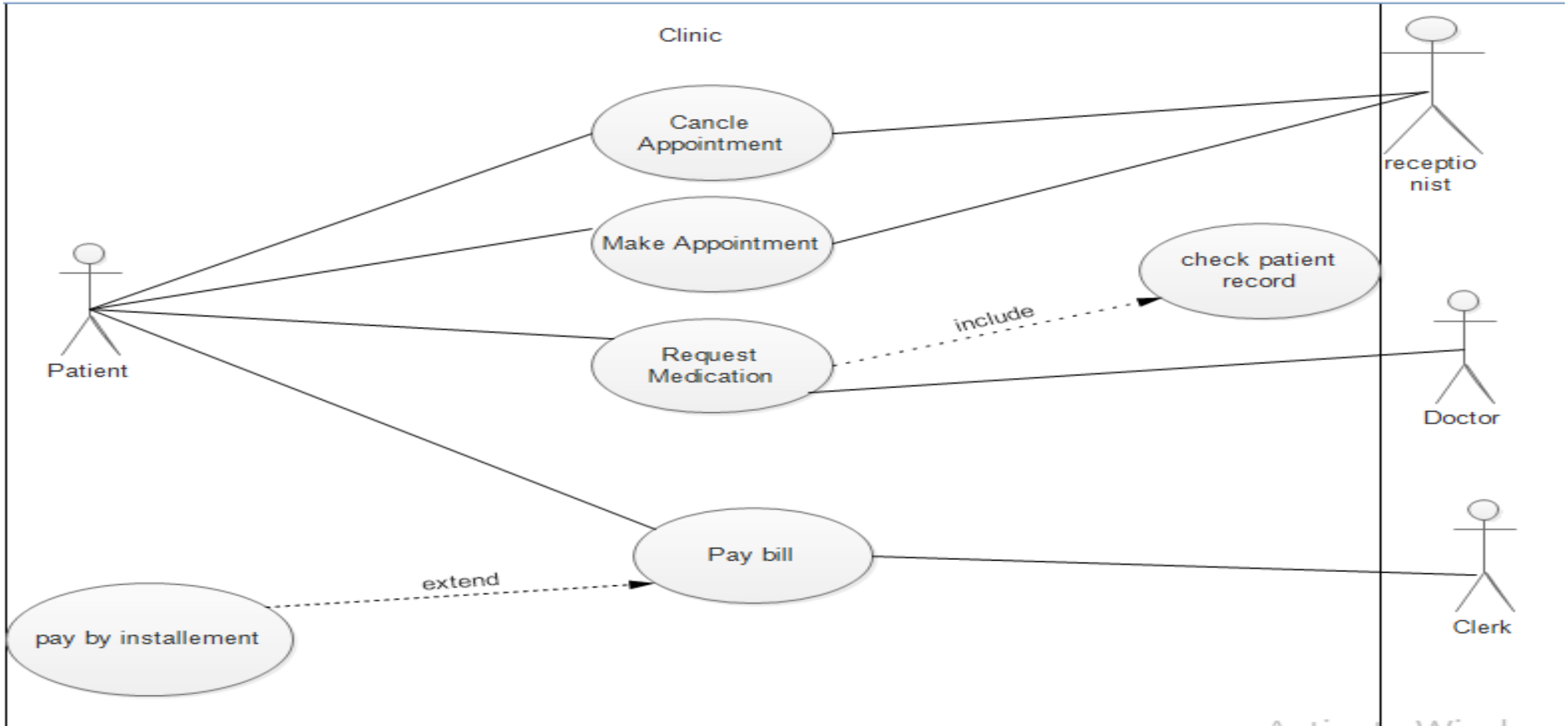
Use Case ID	1
Use Case Name	Login
Actors	Admin,student,liberarian,staff
Preconditions	• Admin,student ,librarian must do registration
Post conditions	login successfully
Normal Flow	1- Insert user name 2- Insert Password 3- Login
Alternative Flows	in step 1 : insert the incorrect user name in step 2 : insert the incorrect password

Use Case#3	Search
Primary Actors	Administrator , librarian, faculty member, Students
Pre-condition	The book or student to searched should have been registered in the database of the library management system
Main Scenario	1. The student or staff or librarian or administrator logins to the system 2. If the login is successful. student or staff or librarian or administrator enters the book name or ISBN or author name and presses search 3. 3.If the search is successful then that book is displayed on the screen. 4. 4.To search for a student the administrator or staff logins to the system. 5. 5.If the login is successful then it is possible to search for any student by entering his id. 6. 6.To search for a staff member the administrator enters his login id. 7. 7.If successful he can search for any of the staff members.
Alternate Scenarios	1.The login fails. 2.The student or librarian or staff or administrator can re-register themselves 3.If the search is unsuccessful then the administrator should add that members .4.If the book search is unsuccessful then that book should be added.

Restaurant use case



Clinic use case



Thank you